MIL-P-50014C(MU)
15 April 1969
SUPERSEDING
MIL-P-50014B(MU)
8 April 1964

MILITARY SPECIFICATION

PACKET, SODIUM PYROPHOSPHATE PEROXIDE

1. SCOPE

1.1 Scope. This specification covers packets containing sodium pyrophosphate peroxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
PPP-B-636 - Box, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

B5-77-209 - Packet, White, Filled, Assembly.
B5-77-485 - Vial, Packet Filling.

FSC 6665

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Sodium pyrophosphate peroxide. The sodium pyrophosphate peroxide shall be a white powder conforming to the requirements in table I when tested as specified therein.

Table I. Chemical and physical requirements

Property	Percent	Test paragraph
Passing 1410-micron sieve (No. 14)	100	4.2.4.1
Titratable hydrogen peroxide, minimum:		
Initial	18.0	4.2.4.2.1
After humidity test	10.0	4.2.4.2.2
After aging	10.0	4.2.4.2.1

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3.2 Manufacture and filling. The packet shall be manufactured as specified on Drawing B5-77-209 and filled with 0.20 ± 0.02 gram (g) of sodium pyrophosphate peroxide conforming to 3.1.1 when tested as specified in 4.2.4.2.4.

3.3 Workmanship. The packet shall be free from damage (tears, slits, holes) and contamination (dirt, foreign matter).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the packets of sodium pyrophosphate peroxide that have been produced by one manufacturer and offered for acceptance at one time. A lot shall contain no more than one batch or sodium pyrophosphate peroxide (see 6.4).

4.2.2 Sampling.

4.2.2.1 For examination. Sampling of packets of sodium pyrophosphate peroxide shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. Thirty packets shall be taken at random from each lot for test.

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4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample packets of sodium pyrophosphate peroxide shall be examined in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For destructive tests. Sample packets of sodium pyrophosphate peroxide shall be tested in accordance with 4.2.4. The acceptance number is 0.

4.2.3.3 Classification of defects.

(a) Packet, white, filled, assembly (Dwg B5-77-209)

Categories	Defects
Critical	None defined
Major:	AQL 1.0 percent defective
101	Filling missing of incomplete
102	Packets incorrectly sealed
103	Container: incorrect: damaged
104	Incorrect quantity of packets
105	Incorrect quantity of packages per pack
106	Marking missing, incorrect, incomplete illegible, non-durable
107	Workmanship (See 3.3)
Minor:	AQL 2.5 percent defective
201	Length or diameter incorrect

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Particle size. Empty the contents of 10 packets into a 1410-micron sieve, conforming to PP-S-366 and assembled with a bottom pan and a top cover. Shake for 10 minutes by hand or for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations per minute and 150 ± 10 taps of the striker per minute.

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4.2.4.2 Titratable hydrogen peroxide.

4.2.4.2.1 Initial content. The hydrogen peroxide content shall be determined for each of 5 packets as follows: Weigh the packet to the nearest 0.0001 g, empty and dissolve the powder in 10 milliliters (ml) of water in a 500-ml iodine flask, and reweigh the empty packet. To the flask, add 30 ml of 30-percent sulfuric acid and 3 drops of saturated ammonium molybdate solution. Mix thoroughly and add, with agitation, approximately 200 ml of 1-percent potassium iodide. Allow the mixture to stand in the closed flask for 5 minutes and titrate the liberated iodine with standardized 0.1N sodium thiosulfate solution, using starch as the indicator. Calculate the percent hydrogen peroxide as follows:

$$\text{Percent hydrogen peroxide} = \frac{1.701\ AB}{C}$$

where: A = milliliters of sodium thiosulfate solution,
B = normality of sodium thiosulfate solution, and
C = grams of sample.

4.2.4.2.2 After humidity test. Place five packets in a desiccator containing water. Packets shall not be in contact with the water. Cover the desiccator and heat in a constant temperature oven at 65° ± 2° C for 8 hours. Determine the percent hydrogen peroxide content of each packet as specified in 4.2.4.2.1.

4.2.4.2.3 After aging. Place 5 packets in an open container and store in a constant temperature oven at 65° ± 2° C for 14 days. Determine the percent hydrogen peroxide content of each packet as specified in 4.2.4.2.1.

4.2.4.2.4 Weight. Weigh the contents of each of 5 packets to determine compliance with 3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, interplant shipment (see 6.6). Fourteen packets of sodium pyrophosphate peroxide prepared as shown on Drawing 85-77-209 shall be packaged in a vial conforming to Drawing 85-77-485, and the cover shall be securely closed.

5.2 Packing, interplant shipment. Unless otherwise specified (see 6.2), 100 filled vials of 5.1 shall be packed upright in a snug-fitting fiberboard box conforming to the minimum requirements of a style RSC, type CF, class domestic variety SW, grade 125 of PPP-B-636. Fiberboard sheets formed from the same materials as the box shall be inserted where needed to form a tight pack. The filled box shall be closed method I or II as specified in the appendix to PPP-B-636.

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5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129. Each package shall also be marked with the following:

Item name.
Lot number.
Number of packets.
Date packaged.
Expiration date (3 years from date of manufacture).

6. NOTES

6.1 Intended use. The packets of sodium pyrophosphate peroxide covered by this specification are intended to be used in detector kits for the detection of G-agents.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of filled vials per shipping container.

6.3 Precautions. Sodium pyrophosphate peroxide will decompose under the influence of metal catalysts (such as iron, copper, lead, magnesium, and cobalt, and their ions). Easily oxidizable or combustible substances will also cause decomposition, often violent. Therefore, contamination by potential catalysts must be avoided when storing, handling, or using this material. Equipment used must be dry: ceramic material, stainless steel, aluminum, well-scoured wood, or fiberboard.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Significant places. For the purpose of determining conformance with this specificaiton, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.6 Interplant shipment. Packaging and packing for supplies and materials that do not directly enter the military supply system. Typical are shipments from a vendor to a contractor, a subcontractor, or a supplier to a military arsenal or plant.

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6.7 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

International interest (see section 6)

Preparing activity:

Army - MU(EA)

Project No. 6665-A241

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel (either Government or contractor) involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Identification numbers on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER(S)

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (If "Yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) (Optional) | DATE

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